BIOINFORMATICS

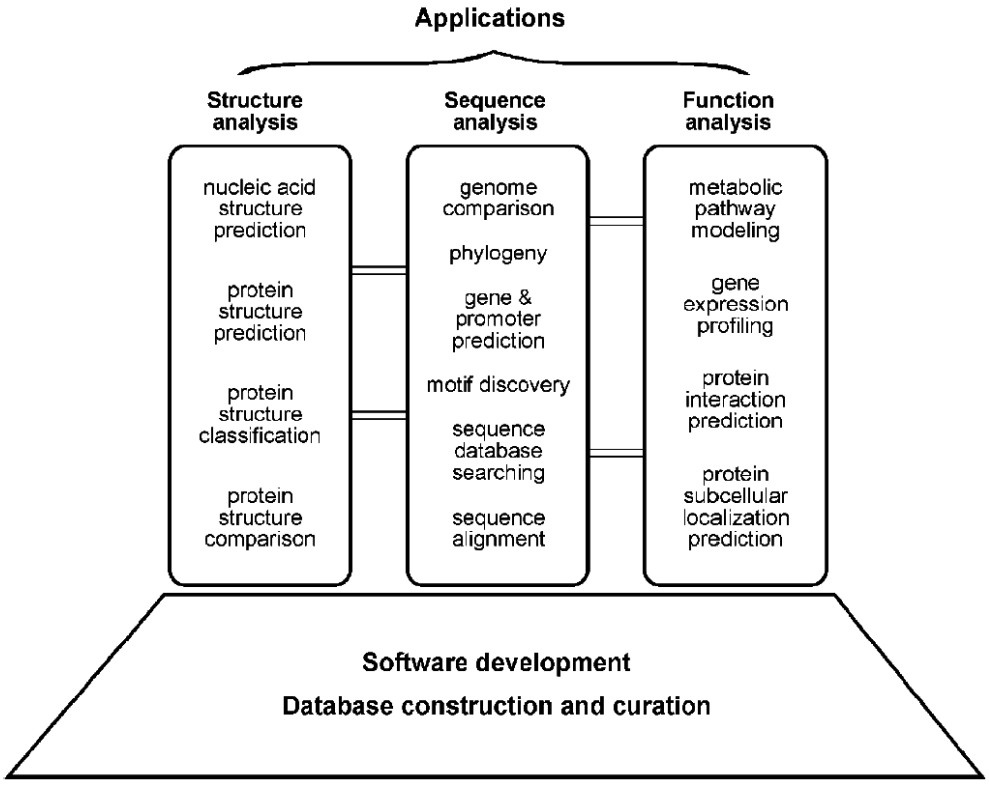


Figure 1.1: Overview of various subfields of bioinformatics. Biocomputing tool development is at the foundation of all bioinformatics analysis. The applications of the tools fall into three areas: sequence analysis, structure analysis, and function analysis. There are intrinsic connections between different areas of analyses represented by bars between the boxes.

What is a Database?

A *database* is a computerized archive used to store and organize data in such a way that information can be retrieved easily via a variety of search criteria. Databases are composed of computer hardware and software for data management.

Based on their contents,

Primary databases, secondary databases, and specialized databases.

Primary databases contain

original biological data , Redundant

archives of raw sequence or structural data submitted by scientific community

GenBank, EMBL and DDBJ are primary sequence databases and Protein Data Bank (PDB) – structural database are examples of primary databases

Secondary databases contain

Computationally processed or manually curated information, based on original information from primary databases

Non-redundant

Multiple entries of the same sequence in primary databases are merged to create a single sequence in the secondary database with extensive annotation derived from all available information on the sequence. The sequence and all the information about it are manually curated.

NCBI RefSeq database is a secondary database that is a collection of curated, non-redundant, well-annotated sequences including genomic DNA, transcripts, and proteins

Table 5.2: Publicly Available Secondary and Specialized Database

Database	**Comments (with URLs)**
Universal Protein Resource Knowledgebase (UniProtKB)	The UniProt Knowledgebase (UniProtKB) is the central repository for the collection of sequence and functional information on proteins with accurate, consistent, and rich annotation. UniProtKB is the product of UniProt, which is an international consortium between the European Bioinformatics Institute (EBI), the Swiss Institute of Bioinformatics (SIB) and the Protein Information Resource (PIR) at the Georgetown University Medical Center. In 2002, EBI, SIB, and PIR started collaboration to create a single high-quality database of protein sequence and function, by unifying the Swiss-Prot, TrEMBL, and PIR-PSO database. Before this collaboration, EMBL-EBI maintained TrEMBL, SIB maintained Swiss-Prot, and PIR maintained the Protein Sequence Database (PIR-PSD). These data sets conexisted with different protein-sequence coverage and annotation priorities[26,27]. (www.uniprot.org) UniProtKB has two sections: UniProt/Swiss-Prot and UniProt/TrEMBL. UniProt/Swiss-Prot contains sequences that are manually annotated, compared, and verified (curated) based on information from literature and curator-evaluated computational analysis. UniProt/TrEMBL (TrEMBL = translated EMBL) contains computationally annotated, unreviewed sequences. TrEMBL sequences are eventually manually curated to become part of Swiss-Prot and removed from TrEMBL.

	Before becoming part of UniProt, PIR-PSD was the oldest annotated and curated protein-sequence database, established in 1984 as a successor to the original National Biomedical Research Foundation (NBRF) Protein Sequence Database. It was developed over a 20 year period by the late Margret Dayhoff and published as the "Atlas of Protein Sequence and Structure" from 1965 to 1978. The link to PIR-PSD is http://pir.georgetown.edu/[28].
Worldwide Protein Data Bank (wwPDB)	Experimentally determined structures of proteins, and complex assemblies. wwPDB is a publicly available archive of macromolecular structural data[29]. (http://www.wwpdb.org/)
Structural Classification of Proteins (SCOP) database	The SCOP database aims to provide a detailed and comprehensive description of the structural and evolutionary relationships between all proteins whose structure is known, including all entries in the PDB, Proteins are classified into families (clear evolutionarily relationship; this generally means that pairwise residue identities between the proteins are 30% and greater), superfamilies (probable common evolutionary origin), and folds (major structural similarity)[30] (http://scop.mrc-lmb.cam.ac.uk/scop/)
Class, Architecture, Topology, Homology (CATH) database	CATH is a manually curated classification of protein domain structures. Each protein is chopped into structural domains and assigned into homologous superfamilies (groups of domains that are related by evolution). This classification procedure uses a combination of automated and manual techniques, which include computational algorithms, empirical and statistical evidence, literature review, and expert analysis[31] (http://www.cathdb.info/)
PROSITE database	This consists of a large collection of biologically meaningful signature patterns or profiles. These signatures are not easily revealed by standard sequence alignment. Each signature can be linked to useful biological information on the protein family, domain, or functional site. Therefore, the database can be used to rapidly and reliably identify which known family of protein (in any) the new sequence belongs to. The PROSITE database uses two kinds of signatures, patterns and generalized profiles, to identify conserved regions[32]

	(http://prosite.expasy.org/)
PRINTS database	This is a compendium of protein fingerprints; a fingerprint is a group of conserved motifs used to characterize a protein family[33] (http://www.bioinf.man.ac.uk/dbbrowser/PRINTS/index.php)
Protein Family (Pfam) database	Pfam is a comprehensive database of protein families; members of a family share significant similarity, thereby suggesting homology. Pfam allows the analys is sequence data in order to search for related proteins in the database based on domains. Domains are regions of the protein, which in different combinations can determine the protein's function. Thus, proteins can be viewed as built from a specific combination of domains. Pfam contains two types of families: high-quality manually curated Pfam-A families and automatically generated Pfam-B families. Pfam uses multiple sequence alignments and hidden Markov models (HMM)[34] (http://www.sanger.ac.uk/resources/databases/pfam.html)
InterPro database	InterPro integrates various predictive protein signatures from diverse source repositories, such as (Gene3D, PANTHER, Pfam, PIRSF, PRINTS, ProDom, PROSITE, SMART, SUPERFAMILY, and TIGRF AMs. Protein signatures from various databases are integrated into InterPro manually. Curators combine signatures representing the same protein family, domain, or site into single database entries, and, where possible, trace biological relationships between the constituent signatures[35] (http://www.ebi.ac.uk/interpro/)
Biological General Repository for Interaction Datasets **(BioGRID)**	The BioGRID database is an online repository of interactions in which data are curated from both high-throughput data sets and individual focused studies, as derived from over 40,000 publications in the primary literature. The current compilation (as of July, 2013) has more than 700,000 raw protein and manually annotated genetic interactions from major model organisms. All BioGRID interaction records are directly mapped to experimental evidence in the supporting publication[36]

	(http://thebiogrid.org/)
Molecular Interaction database (MINT)	MINT is a public repository for protein-protein interactions reported in peer-reviewed journals. It focuses on experimentally verified protein-protein interactions mined from the scientific literature by expert curators. Currently it contains over 240,000 interaction data captured from over 4750 publications[37,38] (http://mint.bio.uniroma2.it/mint/)
Munich Information System for Protein Sequences (MIPS) database	The MIPS mammalian protein-protein interaction database is a resource of high-quality experimental protein-interaction data. The content is based on published experimental evidence that has been processed by human expert curators. MIPS also contains large-scale secondary data of protein similarities, currently containing 38 million non-redundant protein sequences [39, 40] (http://mips.helmholtz-muenchen.de/proj/ppi/)
IntAct	IntAct is a freely available, open source molecular interaction database populated by data either curated from the literature or from direct data depositions. As of September 2011, IntAct contained approximately 275,000 curated binary interaction evidence records from over 5000 publications. The IntAct database also captures protein-small molecule (including phospholipids), protein-nucleic acid, and protein-gene locus interactions[41] (http://www.ebi.ac.uk/intact/)
Structural Database of Allergenic Proteins (SDAP)	SDAP is a web server that integrates a database of allergenic proteins with various computational tools that can assist structural biology studies related to allergens, including predicting the IgE-binding potential of food proteins. This database allows bioinformatics analysis as recommended by the *Codex Alimentarius* and UN Food and Agriculture Organization (FAO)/World Health Organization (WHO) Expert Committee on potential allergenicity of foods derivated through modern biotechnology[a] (http://fermi.utmb.edu/SDAP/)
AllergenOnline/ FARRP database	AllergenOnline provides access to a peer-reviewed allergen list and sequence searchable database intended for the identification of proteins,

(FARRP = Food Allergy Research and Resource Program at the University of Nebraska-Lincoln)	including food proteins, that may present a potential risk of allergenic cross-reactivity. The objective is to identify proteins that may require additional tests, such as serum IgE binding, basophil histamine release, or in vivo challenge to evaluate potential cross-reactivity. (http://www.allergenonline.org/)
Allermatch database	The Allermatch database allows the comparison of a protein sequence with sequences of allergenic proteins in the database, in order to predict whether the protein being evaluated can be allergenic. This database allows bioinformatics analysis as recommended by the *Codex Alimentarius* and FAO/WHO Expert Committee on potential allergenicity of foods derived through modern biotechnology[42] (http://www.allermatch.org/)
Online Mendelian Inheritance in Man (OMIM) database	OMIM is a comprehensive compendium of human genes and genetic-disease-associated phenotypes. The full-text referenced overviews in OMIM contain information on all known Mendelian disorders and over 12,000 genes[b] (http://www.ncbi.nlm.nih.gov/omim/ and http://omim.org/)
ArrayExpress database	A public database of microarray gene-expression data at the EBI. It accepts data generated by sequencing or array-based technologies and currently contains data from almost a million assays, from over 30,000 experiments. Experiments are submitted directly to ArrayExpress or are imported from the NCBI GEO database[43]. ArrayExpress uses the minimum information about a microarray experiment (MIAME) annotation standard[c]. (http://www.ebi.ac.uk/arrayexpress/)
Gene Expression Omnibus (GEO) database	The GEO is a public repository that archives and freely distributes MIAME-compliant microarray data, next-generation sequencing data, and other forms of high-throughout functional genomic data submitted by the scientific community. It is one of three international functional genomics public data repositories, alongside ArrayExpress at the EBI and the DDBJ Omics Archive[44,45] (http://www.ncbi.nlm.nih.gov/geo/)

Array Track database	A public database of microarray gene-expression data at the US Food and Drug Administration. ArrayTrack provides an integrated solution for managing, analyzing, and interpreting microarray gene-expression data and experimental parameters associated with pharmacogenomics or toxicogenomic studies-that is, studies on the effects of drugs or other chemicals on gene expression. ArrayTrack supports MIAME-compliant data[46] (http://www.fda.gov/ScienceResearch/BioinformaticsTools/ Arraytrack/default.htm)
Comparative Toxicogenomic database (CTD)	This is a public database of information built on curated data from the scientific literature about interactions between environmental chemicals and gene products and their relationships to diseases. As of 2013, CTD contains over 15 million toxicogenomic relationships. A user can look up specific literature-based information about genes, gene products, and toxicants of interest and their interactions[47] (http://ctdbase.org/)
Chemical Effects in Biological Systems (CEBS) database	The CEBS database has been developed by the National Center for Toxicogenomics within the National Institute of Environmental Health Science (NIEHS). CEBS integrates data obtained using 'omics technologies (transcriptomics, proteomics, metabolomics) as well as from traditional toxicology studies. Thus, CEBS combines the molecular genetic data with traditional clinical chemistry and histopathology data. This combination allows researchers to fully capture information on dose response, time response, and environmental-stress-induced gene expression. The database captures information from multiple species, such as humans, rats, mice, and *Caenorhabdit is elegans*[48] (http://www.niehs.nih.gov/research/resources/databases/cebs/index.cfm)
DrugMatrix database	DrugMatrix is a toxicogenomic and molecular toxicology database and informatics system developed by the National Toxicology Program (NTP). It contains data from standard toxicological experiments along with large-scale gene-expression data from various organs and tissues. DrugMatrix contains toxicogenomic profiles for 638 different compounds that include approved drugs, withdrawn drugs, and industrial and environmental toxicants[d]

	(https://ntp.niehs.nih.gov/drugmatrix/index.html)
FlyBase database	FlyBase is the leading database and web portal for genetic and genomic information focusing on *Drosophila melanogaster*, but also including data on other *Drosophila* species and related drosophilids. The current content of FlyBase comprises > 200,000 references, including > 87,000 research papers from > 2400 different journals, with publication dates ranging from the seventeenth century through to the present day[49, 50] (http://flybase.org/)
NCBI database	Collection of various databases. This is separately discussed below, in Section 5.6.1 (http://www.ncbi.nlm.nih.gov/)

The BLAT (BLAST-like Alignment Tool) was written by Jim Kent at UCSC. BLAT is used to map the input sequence to the genome - that is, to identify the location of a sequence in the genome. Therefore, BLAT works with the genomic context in memory, but it works by alignment-based similarity search

BLAT works for both DNA and proteins.

BLAT is different from BLAST because, unlike BLAST, BLAT does not search the sequences from GenBank/EMBL-Bank/DDBJ; rather, BLAT uses an index derived from the genome assembly and it consists of all non-overlapping 11-mers except the heavily repeated sequences. For proteins, BLAT uses 4-mers.

SEQUENCE ANALYSIS

Sequence identity means the same residues being present at corresponding positions in two sequences being compared. For proteins, it means the same amino acids; for nucleic acids, it means the same bases.

Sequence similarity means similar residues being present at corresponding positions in the two sequences being compared. For nucleic acids, sequence similarity and sequence identity are the same. However, for proteins, sequence similarity involves amino acids with similar physicochemical and functional properties.

For example,

Substitution of lysine and arginine by one another will be regarded as similar substitution because both are positively charged hydrophilic amino acids.

Substitution of aspartic acid and glutamic acid

Substitution of asparagine by aspartic acid and substitution of glutamine by glutamic acid, or vice versa, are also regarded as similar substitutions.
Substitution of isoleucine, leucine, and valine by one another will be regarded as similar substitutions because they have similar aliphatic hydrophobic side chains.

Substitution of serine and threonine by one another is also regarded as similar substitution.

Similar substitutions are also referred to as conservative substitutions. A conservative amino acid substitution is not expected to disrupt the structural/functional attributes of the protein.

Sequence homology is an evolutionary term. Sequences are called homologous if they have a common evolutionary origin - that is, if they are derived from a common ancestral sequence. So, sequences are either homologous or not homologous and there is no quantitation of homology.

Homologous – genes or proteins derived from a common ancestor, and have not changed much, probably because of functional constraints; hence, they are homologous.

Homologous genes in different species performing the same function are called **orthologs**.

Two proteins can be considered homologous despite low similarity if one or more of the following conditions are met:
(1) the similarity extends over a long stretch of sequence and is statistically significant;
(2) despite low sequence similarity, the same pattern of identical and similar amino acid residues is seen in multiple sequences; or
(3) the pattern of sequence similarity reflects the similarity between experimentally determined structures of the respective proteins, or at least corresponds to the known key elements of one such structure

Phenylalanine and tryptophan have very hydrophobic aromatic side chains, whereas tyrosine and histidine are less so. The latter two can often be found in positions that are somewhere between buried and exposed. The hydrophobic aromatic amino acids can sometimes substitute for aliphatic residues of a similar size, for example phenylalanine to leucine, but not tryptophan to valine.

Sequence identity and sequence similarity can be calculated based on the proportion of identical and similar amino acids, respectively:

% Identity (PID) = (# of identical amino acids / # of total amino acids) × 100; (6.1)

% Similarity = {(# of identical amino acids + # of similar substitutions) / # of total amino acids} × 100 (6.2)

In the above formulae, the denominator (# of total amino acids) can vary. For example, the denominator could be (1) the length of the shortest sequence, (2) the length of the longest sequence, (3) the mean length of the sequences, (4) the length of the aligned region (aligned positions excluding overhangs), etc. Therefore, PID is a rough measure and can be influenced

by how it is calculated. However, because of the simplicity of calculation, PID is widely used.[3]

different. In a protein sequence alignment, *sequence identity* refers to the percentage of matches of the same amino acid residues between two aligned sequences. *Similarity* refers to the percentage of aligned residues that have similar physicochemical characteristics and can be more readily substituted for each other.

There are two ways to calculate the sequence similarity / identity. One involves the use of the overall sequence lengths of both sequences; the other normalizes by the size of the shorter sequence. The first method uses the following formula:

$$S = [(L_s \times 2) / (L_a + L_b)] \times 100 \quad \text{(Eq. 3.1)}$$

where S is the percentage sequence similarity, L_s is the number of aligned residues with similar characteristics, and L_a and L_b are the total lengths of each individual sequence. The sequence identity (*I%*) can be calculated in a similar fashion:

$$I = [(L_i \times 2) / (L_a + L_b)] \times 100 \quad \text{(Eq. 3.2)}$$

where L_i is the number of aligned identical residues.

The second method of calculation is to derive the percentage of identical / similar residues over the full length of the smaller sequences using the formula:

$$I(S)\% = L_{i(s)} / L_a\% \quad \text{(Eq. 3.3)}$$

where L_a is the length of the shorter of the two sequences.

Sequence alignment

The purpose of alignment is to identify regions of similarity that may have structural, functional, and evolutionary consequences

Some widely used online pairwise alignment tools use local alignment strategy (**Smith-Waterman algorithm)** and are shown in Table 6.1.

Table 6.1: Online Pairwise Alignment Tools Using the Smith – Waterman Algorithm

Online Tool	**URL**
PIR Research	http://pir.georgetown.edu/pirwww/search/pairwise.shtml[5]
NCBI specialized BLAST	bl2seq resource; look for the Align link on the NCBI BLAST home page under Specialized BLAST

SIM	http://web.expasy.org/sim/
LALIGN*	http://www.ch.embnet.org/software/LALIGN_form.html

The LALIGN program is William Pearson's, and it implements the algorithm of X. Huang and W. Miller.[6]

Multiple sequence alignments are useful in identifying conserved sequence segments across the sequences being aligned.

Such **conserved regions across multiple sequences usually indicate an evolutionary relationship**. For an unknown protein, for example, such conserved sequence segments identified through multiple alignment can be used in conjunction with other information to predict functionally important and evolutionarily conserved motifs within the proteins

MSA is also needed for construction of phylogenetic trees.

Most widely used MSA tool is ClustalW or Clustal Omega

The T-Coffee, CBRC (Computational Biology Research Center at the National Institute of Advanced Industrial Science and Technology, Japan) MAFFT, and EMBL-EBI MUSCLE all use ClustalW, so the output format is similar. NCBI COBALT has a very different output format.

In order to perform multiple alignment, the ClustalW algorithm goes through a number of steps, as follows:

it calculates all possible pairwise alignments of the input sequences;
computes the score of each alignment, where the score reflects the distance between the two sequences;
creates a dendrogram (guide tree) based on the matrix of the distance;

and uses the dendrogram as the basis to perform multiple alignment, where closely related pairs of sequences are aligned first.

Table 6.2: Online Multiple Alignment Tools

Online Tool	**URL**
COBALT (NCBI)	http://www.ncbi.nlm.nih.gov/tools/cobalt/cobalt.cgi?link_loc = BlastHomeLink[7]
ClustalW (DDBJ)	http://clustalw.ddbj.nig.ac.jp/inde.php?lang=en[8, 9]
MAFFT (CBRC)	http://mafft.cbrc.jp/alignment/server/[10]

MUSCLE (EMBL-EBI)	http://www.ebi.ac.uk/Tools/msa/muscle/[11]
T-Coffee	http://www.tcoffee.org/Projects/tcoffee/. Then click any of the server links on this page, such as http://www.tcoffee.org/ and from there the type of alignment program needed for analysis[12]

ALIGNMENT ALGORITHMS, GAPS, AND GAP PENALTIES

An algorithm is a step-by-step procedure that utilizes a finite number of instructions for automated reasoning and the calculation of a function.

The algorithm that drives **global alignment** is the **Needleman- Wunsch algorithm**, and the algorithm that drives **local alignment** is the **Smith-Waterman algorithm**.

Both these algorithms are examples of dynamic programming.

Dynamic programming is a method for solving complex problems by breaking them down into simpler sub problems.
In the case of sequence alignment, dynamic programming involves setting up a two-dimensional matrix in which one sequence is listed vertically and the other sequence is listed horizontally; then calculating the scores, one row at a time. For example, a match can be given a 1, a mismatch a 0, and a gap a 21. A 100% perfect alignment will produce a diagonal straight line (with a negative slope) spanning from the top left to bottom right. If the alignment is not perfect, gaps are introduced in the matrix. For the sequence represented horizontally, gaps are introduced vertically, and for the sequence represented vertically, gaps are introduced horizontally, and the alignment is determined by a traceback step.

The basic sequence alignment method is the dot matrix or dot plot method. In this method, two sequences being compared are written in the vertical and horizontal axes of the matrix. Then each residue is scanned and each match is given a dot; mismatches are left blank. When enough dots are lined up, they are connected.

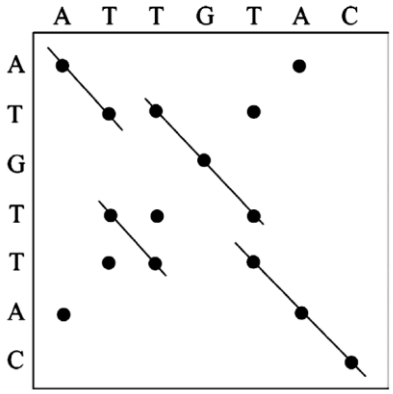


Figure 6.8 Comparison of two sequences using dot matrix or dot plot.

In both global and local alignment, the final output is given an alignment score. **Gaps have to be introduced to improve the alignment. The reason gaps are introduced** is because one of the sequences may have **gained or lost sequence characteristics** (insertion————————————deletion) during evolution that did not happen with the other sequence.

However, the **number of gaps is kept to a minimum to keep the alignment meaningful**; otherwise an artificially **high alignment score can be obtained even when the two sequences are not related.** The gap penalty value is subtracted from the gross alignment score to obtain the final alignment score (alignment score and scoring matrix are discussed in the next section). The insertion of no more than 1 gap per 20 amino acid residues is ideal but that is not possible in most cases. **For each gap opened, a gap-opening penalty** value is assigned, and **for each gap extended, a gapextension penalty value is assigned.** A **gap-opening penalty is always much higher than a gap-extension penalty**. Often, a default value of -10 for a gapopening penalty and -1 for a gap-extension penalty are used. However, these values can be different and can also be adjusted by the user. This type of differential penalty for gap opening and gap extension is called affine gap penalty.

There are other types of gap penalties, such as constant gap penalty, linear gap penalty, and proportional gap penalty, but for all practical purposes affine gap penalty is the most relevant for sequence alignment. **Affine gap penalty is calculated as follows**:

$$G_t = G_o + G_e \times L_n, \quad (6.3)$$

where G_t = total gap penalty, G_o = gap – opening penalty, G_e = gap – extension penalty, and L_n = length of the extension gaps. For any given block of gaps, L_n = # of total gaps – 1, because the first gap is the opening, the rest in the block are extensions.

A raw alignment score can be calculate based on the following simple formula:

$$S = \sum_i + \sum_m - G_t, \quad (6.4)$$

where S = raw score, $\sum_i$ = total score for identities, $\sum_m$ = total score for mismatches, and G_t = total gap penalty.

For both nucleic acids and proteins, the **alignment score is calculated using a scoring matrix**. **A scoring matrix is a set of values representing the likelihood of one residue being substituted by another during sequence divergence through evolution**.

This is why the scoring matrix is also known as the substitution matrix.

A scoring matrix for comparing DNA sequences can be simple because there are only four nucleotides and the mutation frequencies are assumed to be equal (the Jukes and Cantor assumption). A high positive score (e.g. 5) is assigned for a match and a low negative score (e.g. 24) for a mismatch, thus creating a simple model.

However, the **frequency of transition mutations** (purine replaced by purine or pyrimidine replaced by pyrimidine) **is higher than transversion mutations** (purine replaced by

pyrimidine or vice versa). To deal with this differential mutation frequency, sophisticated statistical models have been developed by Kimura and others. For generating a DNA sequence-alignment score, the simple scoring matrix is still used, such as the NUC4.2 and NUC4.4 DNA scoring matrices. These matrices can be obtained from the NCBI (ftp://ftp.ncbi.nih.gov/blast/matrices/).

Scoring matrices for amino-acid substitutions are more complex, reflecting the similarity of physicochemical properties, as well as the likelihood of one amino acid being substituted by another at a particular position in homologous proteins. The scoring matrices for proteins are 20 X 20 matrices. **Two well-known types of scoring matrices for proteins are PAM and BLOSUM.**

PAM (point accepted mutation—that is, accepted point mutation—also called percent accepted mutation) matrices were first **developed by Margaret Dayhoff** and colleagues in 1978 and hence are **also known as Dayhoff PAM matrices**.

A PAM represents a substitution of one amino acid by another that has been fixed by natural selection because either it does not alter the protein function or it is beneficial to the organism.

In a PAM1 matrix, which is the original PAM matrix generated, a PAM unit is an evolutionary time over which 1% of the amino acids in a sequence are expected to undergo accepted mutations, resulting in 1% sequence divergence.

The PAM1 matrix was built by aligning closely related protein sequences (71 protein families) that had at least 85% sequence identity.

Subsequently, in order to deal with protein sequences that are more diverged and distantly related, other PAM matrices, such as PAM100 and PAM250, were generated. These later PAM matrices were generated by multiplying the PAM1 matrix by itself hundreds of times. For example, the **PAM250 matrix can be obtained by multiplying the PAM1 matrix by itself 250 times over.** Figure 6.9 shows the PAM250 substitution matrix. The values in the matrix are log odds scores

PAM 250 Matrix

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
Ala A	2																			
Arg R	-2	6																		
Asn N	0	0	2																	
Asp D	0	-1	2	4																
Cys C	-2	-4	-4	-5	12															
Gln Q	0	1	1	2	-5	4														
Glu E	0	-1	1	3	-5	2	4													
Gly G	1	-3	0	1	-3	-1	0	5												
His H	-1	2	2	1	-3	3	1	-2	6											
Ile I	-1	-2	-2	-2	-2	-2	-2	-3	-2	5										
Leu L	-2	-3	-3	-4	-6	-2	-3	-4	-2	2	6									
Lys K	-1	3	1	0	-5	1	0	-2	0	-2	-3	5								
Met M	-1	0	-2	-3	-5	-1	-2	-3	-2	2	4	0	6							
Phe F	-3	-4	-3	-6	-4	-5	-5	-5	-2	1	2	-5	0	9						
Pro P	1	0	0	-1	-3	0	-1	0	0	-2	-3	-1	-2	-5	6					
Ser S	1	0	1	0	0	-1	0	1	-1	-1	-3	0	-2	-3	1	2				
Thr T	1	-1	0	0	-2	-1	0	0	-1	0	-2	0	-1	-3	0	1	3			
Trp W	-6	2	-4	-7	-8	-5	-7	-7	-3	-5	-2	-3	-4	0	-6	-2	-5	17		
Tyr Y	-3	-4	-2	-4	0	-4	-4	-5	0	-1	-1	-4	-2	7	-5	-3	-3	0	10	
Val V	0	-2	-2	-2	-2	-2	-2	-1	-2	4	2	-2	2	-1	-1	-1	0	-6	-2	4

Figure 6.9 A PAM250 substitution matrix made by writing amino acids in alphabetical order.

BLOSUM will be referred to as BLOSUM matrix here. BLOSUM (blocks substitution matrices) scoring matrices were proposed by Steven Henikoff and Jorja Henikoff in 1992. BLOSUM represents an alternative set of scoring matrices, which are widely used in sequence alignment algorithms.

Like PAM, BLOSUM matrices are also log-odds matrices.

BLOSUM matrices were developed based on multiple alignment of 500 groups of related protein sequences, which yielded.2000 blocks of conserved amino-acid patterns. Blocks are ungapped multiple sequence alignments corresponding to the most conserved regions of the proteins involved. Henikoff and Henikoff used their BLOCKS database of trusted alignments. In each multiple alignment, the sequences showing similar % identity were clustered into groups and averaged. Using these groups, the substitution frequencies for all pairs of amino acids were calculated and the matrix was developed.

Different BLOSUM matrices differ in the % sequence identity used in clustering. Therefore, **BLOSUM62** means that the **sequences used to create this matrix have approximately 62% identity.** Substitution frequencies weigh more heavily by protein sequences having less than 62% identity. **Therefore, BLOSUM62 is useful for aligning and scoring proteins that show less than 62% identity.** Shown below is an example of an ungapped multiple alignment. The conserved amino acids are shaded for identification.

Henikoff and Henikoff tested the performance of hierarchical multiple alignment of three serine proteases using **BLOSUM45, BLOSUM62, BLOSUM80, PAM120, PAM160, and PAM250** matrices. **All BLOSUM matrices performed better than PAM matrices**; the number of residues misaligned was three to five times lower when BLOSUM matrices were used compared to PAM matrices. **BLOSUM62 performed slightly better than BLOSUM45 and BLOSUM80**.

BLOSUM 62 matrix

Ala	A	4																	
Arg	R	-1	5																
Asn	N	-2	0	6															
Asp	D	-2	-2	1	6														
Cys	C	0	-3	-3	-3	9													
Gln	Q	-1	1	0	0	-3	5												
Glu	E	-1	0	0	2	-4	2	5											
Gly	G	0	-2	0	-1	-3	-2	-2	6										
His	H	-2	0	1	-1	-3	0	0	-2	8									
Ile	I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4								
Leu	L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4							
Lys	K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5						
Met	M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5					
Phe	F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6				
Pro	P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7			
Ser	S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4		
Thr	T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5	
Trp	W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11
Tyr	Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7
Val	V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4
		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V

Figure 6.10 BLOSUM62 substitution matrix made by writing the amino acide in alphabetical order.

The following shows the PAM_BLOSUM matrix equivalence, and their preferred use:

PAM 100 ≈ BLOSUM 90	(for less divergent proteins)
PAM 120 ≈ BLOSUM 80	
PAM 160 ≈ BLOSUM 62	(for most other proteins)
PAM 200 ≈ BLOSUM 52	
PAM 250 ≈ BLOSUM 45	(for more divergent proteins)

In general, BLOSUM matrices are widely used for detecting local alignments. BLOSUM62 is the most frequently used matrix for detecting the majority of weak protein similarities, and BLOSUM45 is very suitable for detecting long and weak alignments. While aligning unknown sequences, if one wants to use the most appropriate matrix based on how similar the sequences are, one has to first try multiple matrices and then use the one that gives the highest ungapped alignment score.

Statistical Significance of Sequence Alignment

6.7.3.1 P-Value

The P-value of an alignment represents the probability of obtaining a score ≥ S by chance. For example, if the P-value is 10^{-5}, it means that the probability of obtaining an alignment with a score ≥ S is 1 out of 10^5. Thus, different alignments can be compared based on their P-values. The P-value ranges from 0 to 1; the closer it is to 0, the better is the alignment.

6.7.3.2 Z-Score

In the statistical sense, Z is the distance between S and the mean of scores obtained using randomized sequences.
Z > 20: two sequences are definitely homologous (family)
Z between 10 and 20: two sequences most likely homologous (Family / Superfamily)
Z between 6 and 8: two sequences are less likely to be homologous
Z < 6: not significant.

6.7.3.3 E-Value

This is particularly relevant in relation to sequence-similarity searching using BLAST and FASTA, which are discussed later in this chapter. The **E-value** is the **expectation** value that indicates the number of alignments with a score ≥ S that one can expect to find by chance in a database of size N. Hence, the E-value is dependent on the database size N. Hence, the E-value is dependent on the database size and the query length. The closer the E-value to 0, the better is the alignment. For E < 1e – 2 (= 1×10^{-2} = 0.01), P ≈ E. *The E-value is the most widely used measure for estimating the quality of sequence alignment – that is, the extent of sequence similarity.*

The typical threshold for the E-value when judging homology, particularly using BLAST, is E $\leq$ 1e – 5 (= 1×10^{-5}), and the lower the value, the better it is. For BLAST (both nucleotide and protein), the default E-value is set at 10 in the **Expect threshold** box under **Algorithm parameters** (lower left corner of the BLAST home page). This means that 10 matches are expected to be found merely by chance, according to the stochastic model of Karlin and Altschul (1990).[19]

6.8.2.3 Suggested BLAST E-Value Cut-off

For nucleic-acid-based search, the suggested threshold (minimum significant hit) for the E-value is $\leq$ 1e-6 (= 10^{-6}), and a sequence identity of $\geq$ 70%. For protein-based search, the suggested threshold for the E-value $\leq$ le-4 (= 10^{-4}), with a sequence identity of $\geq$ 35%[g]. However, typically for protein-based homology search the threshold used is E $\leq$ 1e-5 (= 10^{-5}), and the lower it is, the better. For example, an E-value of le-25 (= 10^{-25}) will indicate a clear homology.

6.7.3.4 Bit Score

The **bit score** (S') is a normalized raw score expressed in *bits*; it is an estimate of the search space one has to search through – that is, the number of sequence pairs one has to score – before one can come across a raw alignment score $\geq$ S, by chance.

For example, a bit score of 30 means that, on average, one has to score 2^{30} (= 1 billion) sequence pairs before one will come across a score $\geq$ S, by chance. Usually, good alignments produce a bit score > 50. *It should be emphasized that the bit score is dependent on sequence length, and short sequences may not produce high bit scores despite very high identity.*

To summarize the utility of the statistical estimates of sequence alignment in simple terms, **the better the alignment (e.g. homologous sequences), the lower the P-values and E-values, and the higher the Z-values and bit scores.**

Megablast, Blastn, and Discontinuous Megablast

Megablast is optimized for highly similar sequences. It efficiently finds long alignments between highly similar (>95%) sequences, and thus is the best tool to find the identical match to the query sequence. The default **word size is 28 and the lowest word size is 16**.

Blastn is optimized for somewhat similar sequences. The reason blastn is more sensitive than megablast is because it uses a shorter default **word size (11)**

blastn is better than megablast at finding alignments to related nucleotide sequences from other organisms. Reducing the word size from 11 (default) to 7 (lowest) increases the sensitivity of search—that is, increases the number of positive hits.

Discontinuous megablast is optimized for more dissimilar or diverged sequences especially sequences from different organisms. Instead of using the exact word match as seed for an

alignment extension, discontinuous megablast uses a noncontiguous word within a longer window of template

6.8.3.1 Searching for Distantly Related Proteins – PSI-BLAST

Many homologous proteins have similar three-dimensional structure, but in pairwise alignment they may not show significant sequence similarity. Therefore, regular protein BLAST (blastp) is not useful in identifying these proteins. Position-Specific Iterative BLAST (PSI-BLAST) is designed to detect weak relationships between the query sequence and other sequences in the database that are not necessarily detectable by standard BLAST searches. When a new genome is

6.8.3.2 Searching for Pattern Hit—PHI-BLAST

Many proteins contain signature sequences (motifs) that are characteristics of a protein family. These signature sequences are part of important structural or functional domains. Pattern-hit-initiate (PHI)-BLAST is designed to search the database for proteins that are significantly related to the query sequence and also contain a pattern. In other words, PHI-BLAST searches for significantly similar sequences to both a query sequence and a signature. This dual requirement is supposed to reduce the number of database hits that contain the pattern but are likely to have no true homology to the query.

6.8.5.1 Comparison of BLAST and FASTA

BLAST and FASTA are both heuristic algorithms that perform database searches to find sequences related to a query sequence. However, there are some differences between the two:

1. BLAST begins a search by looking for matches that include exact matches and conservative substitutions; FASTA begins a search by looking at exact matches.

2. BLAST scans a larger window size that FASTA; hence, FASTA may produce better coverage for homologs.

3. BLAST may produce multiple best-scoring alignments (also called **high-scoring segment pairs or HSPs**) from the same sequence; FASTA returns only one alignment from one sequence.

4. BLAST automatically masks low-complexity regions; FASTA does not employ such automatic masking. Therefore, if the query sequence has non-unique segments such as repeats, compositionally biased segments, etc., FASTA search may return alignments with artificially high scores.

5. For a given sequence search, the BLAST output is larger than that of FASTA.

6. For a given sequence search, BLAST is faster than FASTA.

Table 2.1: Major Biological Databases Available Via the World Wide Web

Database and Retrieval System	**Brief Summary of Content**	**URL**
AceDB	Genome database for *Caenorhabditis elegans*	www.acedb.org
DDBJ	Primary nucleotide sequence database in Japan	www.ddbj.nig.ac.jp
EMBL	Primary nucleotide sequence database in Europe	www.ebi.ac.uk/embl/index.html
Entrez	NCBI portal for a variety of biological databases	www.ncbi.nlm.nih.gov/gquery/gquery.fcgi
ExPASY	Proteomics database	http://us.expasy.org/
FlyBase	A database of the *Drosophila* genome	http://flybase.bio.indiana.edu/
FSSP	Protein secondary structures	www.bioinfo.biocenter.helsinki.fi:8080/dali/index.html
GenBank	Primary nucleotide sequence database in NCBI	www.ncbi.nlm.nih.gov/Genbank
HIV databases	HIV sequence data and related immunologic information	www.hiv.lanl.gov/content/index
Microarray gene expression database	DNA microarray data and analysis tools	www.ebi.ac.uk/microarray
OMIM	Genetic information of human diseases	www.ncbi.nlm.nih.gov/entres/query.fcgi?db=OMIM

PIR	Annotated protein sequences	http://pir.georgetown.edu/pirwww/pirhome3.shtml
PubMed	Biomedical literature information	www.ncbi.nlm.nih.gov/PubMed
Ribosomal database project	Ribosomal RNA sequences and phylogenetic trees derived from the sequences	http://rdp.cme.msu.edu/html
SRS	General sequence retrieval system	http://srs6.ebi.ac.uk
SWISS-Prot	Curated protein sequence database	www.ebi.ac.uk/swissprot/access.html
TAIR	Arabidopsis information database	www.arabidopsis.org

Table 8.5: Some Online Tools for the Analysis of Possible Secondary Structure of a Protein

Online Tool	**Comments and URL**
APSSP	http://imtech.res.in/raghava/apssp/*
CFSSP (Chou-Fasman[16] Secondary-Structure Prediction)	http://www.biogem.org/tool/chou-fasman/*
GOR IV	GOR IV[17]; GOR I, the original GOR[18] (http://npsa-pbil.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_gor4.html)*[19]
HMMSTR	HMM-based[20,21] (http://www.bioinfo.rpi.edu/bystrc/hmmstr/server.php)
JPred 3	Combines the analysis from multiple prediction algorithms, such

	as DSC, JNET, PHD, and PREDATOR[22] (http://www.compbio.dundee.ac.uk/www.jpred/)
NPS@ (Network Protein Sequence Analysis	This site contains links to a number of prediction tools including GOR and PHD. However, GOR and PHD are mentioned here separately as well. Pay attention to those that were developed in the late 1990s. Compare the output from these tools[*19] (http://npsa-pbil.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_server.html)
PHD	Neural-network-based[23-25] (http://npsa-pbil.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_phd.html)
PredictProtein	Meta-server that combines the analysis from multiple prediction algorithms such as Jpred, PHD, PROF, and PSIPRED. It is a good secondary-structure prediction program[**] (https://www.predictprotein.org/)
PROTEUS 2	Combination of HMM-and neural-network-based prediction[26] (http://wishart.biology.ualberta.ca/proteus2/)
PSIPRED	Combination of homology modeling and neural-network-based prediction. It is a good secondary-structure prediction program[22] (http://bioinf.cs.ucl.ac.uk/psipred/)
Quick2D	Provides an overview of secondary-structure features like $\alpha-$ helices, extended $\beta-$ sheets, coiled coils, transmembrane helices, and disordered regions. Predictions by PSIPRED, JNET, Prof(Rost), Prof (Ouali), Coils, MemSAT2, HMMTOP, DISOPRED2 and VSL2[++] (http://toolkit.tuebingen.mpg.de/quick2_d)
SCRATCH Protein Predictor	The SCRATCH software suite includes predictors for a number of parameters, such as secondary structure, relative solvent

	accessibility, disordered regions, domains, individual residue contracts, teritiary structure and more[28] (http://scracth –proteomics.ics.uci.edu/index.html)
SSPro 4.0	Bidirectional recurrent neural network (BRNN)-based[29, 30] (http://scratch.proteomics.ics.uci.edu/sspro4.html)
SYMPRED	SYMPRED can be run using any combination of the following programs: PHD, PROF, SSPro201, YASPIN, JNet and PSIPRED. The consensus of the outputs is derived through dynamic programming to achieve a higher level of prediction accuracy[31] (http://wwwibi.vu.nl/programs/sympredwww/)
SOPMA	An improved self-optimized prediction method (SOPM)[32] (http://npsa-pbit.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_sopma.html)
YASPIN	Neural-network-based[33] (http://www.ibi.vu.nl/programs/yaspinwww/)

SYNTENY

A chromosomal segment that has been inherited from the common ancestor during evolution without a major rearrangement of the order of genes is called a **syntenic block** (or synteny block). Syntenic blocks contain specific non-repetitive genomic markers that are in the same order and orientation in the genomes being compared. These genomic markers could be protein-coding genes, RNA-coding genes, noncoding sequences, pseudogenes, etc., and are called **syntenic anchors** (or synteny anchors)

Charting the blocks of conserved synteny creates a **synteny map**, which shows the large-scale evolutionary relationships between genomes that are related through a common ancestor, but have diverged during evolution.

Table 7.1: Some Online Tools for Prediction of Promoters, Cis-Regulatory Elements, Transcription Start and Initiation Sites, and the ORF

Online Analysis Tool	**Comments and URL**

BPROM	**Bacterial promoter prediction.** A SoftBerry utility that predicts putative transcription start positions of bacterial genes regulated by sigma 70 promoters. The prediction accuracy is about 80%; the specificity is also about 80% when tested on equal numbers of promoter and non-promoter sequences. It uses the signal and content information of the sequence (e.g. consensus sequence). BPROM should be run on a region between two neighboring ORFs located on the same strand, or on a sequence upstream from an ORF (most promoters are located within 150 bp upstream of the ORF). BPROM should not be used for whole genomes, to avoid the many false positives (http://linux1.softberry.com/berry.phtml?topic=bprom&group=programs &subgroup=gfindb)
Virtual Footprint	**Prokaryotic promoter prediction.** Virtual Footprint is a software suite for analyzing transcription-factor-binding sites in whole bacterial genomes and their underlying regulatory networks. The results is a list of potential binding sites and corresponding genes defining the whole regulon. There are two types of analysis: analysis of a whole prokaryotic genome with one regulator pattern, and analysis of a promoter region with several regulator patterns[28] (http://www.prodoric.de/vip/vfp_promoter.php)
BDGP (Berkeley *Drosophila* Genome Project)	**Prokaryotic and eukaryotic promoter prediction.** Neural network promoter prediction (NNPP)-based. NNPP is method that consists mainly of two recognition features for predicting eukaryotic promoters; one for recognizing the TATA-box and one for recognizing the initiator element. Both features are combined into one output unit, which gives output scores between 0 and 1. The default score is set at 0.8. The prediction accuracy for prokaryotic promoters is greater than that for eukaryotic promoters[29] (http://www.fruitfly.org/seq_tools/promoter.html)
FindTerm	**Rho-independent-terminator prediction in the bacterial genome.** A SoftBerry utility that predicts terminators in the bacterial genome. The search utilizes certain known features of bacterial terminators, such as T-rich regions, possible combinations of spacer lengths, all hairpins etc., and the result output shows all putative terminators (http://linux1.softberry.com/berry.phtml?topic=findterm&group

	=programs&subgroup=gfindb)
Promoter 2.0	**Vertebrate pol II transcription start site (TSS) prediction.** The program builds on principles that are common to neural networks and genetic algoritms[30] (http://www.cbs.dtu.dk/servies/Promoter/)
Tfsitescan	**Eukaryotic promoter sequence and putative transcription-factor-binding site prediction.** Works best with sequences of ~ 500 nt. The output is in graphic display and shows expectation sores for the putative binding sites[a] (http://www.ifti.org/cgi-bin/ifti/Tfsitescan.pl)
SoftBerry Search for promoters / functional motifs	**SoftBerry utility providing a suite of prediction tools for promoter/ functional motif prediction.** For example: 1. Plant promoter prediction (**TSSP**) 2. Human pol II promoter prediction (**TSSG and TSSW**) 3. Human promoter prediction (**EPROM**) 4. Promoter prediction using orthologous sequences in eukaryotic genome (**PromH(G) and PromH(W)**) 5. Regulatory motif prediction (**Nsite**) (http://linux1.softberry.com/berry.phtml?topic=index&group=programs &subgroup=promoter)
WWW Signal Scan	**Eukaryotic transcriptional elements prediction** based on scoring homologies of published *cis-regulatory* transcriptional signal sequences (e.g. in TFD, TRANSFAC databases) in the input sequences[b,31] (http://www-bimas.cit.nih.gov/molbio/proscan/)
Human Core-Promoter Finder	**Transcription start site (TSS) prediction in human core-promoters.** The input genomic DNA sequence should be longer than 240 bp and less than 2001 bp. The functional core-promoter is assumed to span between – 60 and + 40 nt with respect to the TSS (+ 1). The program is able to localize a TSS to a 100-bp interval ~60% of the time[c].

	(http://rulai.cshl.org/tools/genefinder/CPROMOTER/human.htm)
EP3 (Easy Promoter Prediction Program)	**Eukaryotic core promoter prediction.** Performs very well in identifying regions in human genes that are associated with transcription initiation. EP3 uses universal properties of the promoter to detect those regions in a whole-genome context[33] (downloadable) (http://bioinformatics.psb.ugent.be/webtools/ep[3])
Eponine	**Transcription start site prediction in mammalian genomic sequence.** A probabilistic method with good specificity and excellent positional accuracy. Eponine is estimate to detect > 50% of transcription start sites, with ~70% specificity[34] (downloadable from Sanger Center) (http://www.sanger.ac.uk/resources/software/eponine/)
Footprinter	**Prediction of regulatory elements in DNA sequences based on phylogenetic footprinting.** Phylogenetic footprinting method identifies regions of DNA that are highly conserved across a set of orthologous sequences[35] (downloadable from the University of Washington (Motif Discovery link) (http://bio.cs.washington.edu/software)
ORF Finder	**Open reading frame (ORF) prediction.** A very user-friendly ORF finder on the web. It is a graphical analysis tool that finds all ORFs in the input sequence, using the standard or alternative genetic codes. The putative ORFs are displayed in six frames, three sense and three antisense[d] (http://www.ncbi.nlm.nih.gov/gorf/gorf.html)
NetStart 1.0	**Translation initiation site prediction.** NetStart produces neural network predictions of translation start sites in vertebrate and *Arabidopsis thaliana* nucleotide sequences. The program has been trained on cDNA-like sequences; therefore, it shows better performance for cDNAs and ESTs. It has not been tested on genomic data[36] (http://www.cbs.dtu.dk/services/NetStart/)
ATGPr	**Translation initiation site prediction.** ATGpr can be used to predict whether an initiation codon for cases where there are multiple ATG

	condons. The method uses linear discriminant analysis, and has been tested on a non-redundant data set of 660 sequences[37] (http://atgpr.dbscls.jp/)

Table 7.2: Some Online Tools for RNA Secondary-Structure Prediction

Online Analysis Tool	Comments and URL
RNAfold	RNAfold predicts secondary strcutures of single-stranded RNA or DNA sequences based on the classic minimum-free-energy algorithm of Zuker and Stiegler[42] as well as the partition-function algorithm of McCaskill.[43] Current limits are 10,000 nt for minimum-free-energy-only predictions and 7500 nt for partition-function calculations. The server function can be tested using the sample sequence provided[44] (http://rna.tbi.univie.ac.at/cgi-bin/RNAfold.cgi)
RNAsoft	RNAsofts is a collection of online services for the computational prediction and design of RNA/DNA structures based on a standard free-energy model.[45] The underlying algorithms have been designed and implemented by members of the Bioinformatics, Empirical and Theoretical Algorithmics (BETA) Lab at the Department of Computer Science of the University of British Columbia (http://www.rnasoft.ca/)
CONTRAfold	CONTRAfold is a novel secondary-structure prediction method based on conditional log-linear models (CLLMs), a flexible class of probabilistic models with high prediction accuracy[46] (http://contra.stanford.edu/contrafold/server.html)
RNAstructure	RNAstructure uses several secondary-structure prediction algorithms, including thermodynamic and partition-function algorithms. It is a complete package for RNA and DNA secondary-structure prediction and analysis. It can take different types of experiment mapping data to constrain or restrain structure prediction[47]

	(http://rna.urmc.rochester.edu/RNAstructureWeb/)
IPKnot	IPKnot performs integer-programming (IP)-based prediction of RNA pseudoknots. IPknot can also predict the consensus secondary structure when a multiple alignment of RNA sequences is given[48] (http://rna.naist.jp/jpknot/)
CYLOFOLD	RNA secondary-structure (including pseudoknot) prediction tool. Some examples of RNA sequences are provided that can be used to perform a test run. The bracket notation output is in brackets and hyphens instead of brackets and dots* (http://cylofold.abcc.ncifcrf.gov/)
CentroidHomfold and CentroidFold	CentroidHomfold predicts the secondary structure of an input RNA sequence by employing automatically collected homologous sequences of the target[49,50] CentroidFold uses the CONTRAfold model as the default setting to calculate base-pairing probabilities, and predicts RNA secondary structure using a γ - centroid estimator. Currently, the input sequence should be less than or equal to 200 bases[51] (http://www.ncrna.org/)
pknotsRG	pknotsRG is a tool for prediction RNA secondary structures, including the class of simple recursive pseudoknots. It uses the thermodynamic energy model extended by some pseudoknot-specific values.[52] The program on the BiBiserv is limited to sequences of length up to 800 bases (http://bibiserv.techfak.uni-bielefeld.de/pknotsrg/submission.html) pknotsRG will be discontinued and replaced by pKiss in the near future (http://bibiserv2.cebitec.uni-bielefeld.de/pkiss)

Table 7.3: IUPAC Codes for Nucleotides

A = adenine	**T = thymine**	**G = guanine**	**C = cytosine**		
R = A/G	Y = C/T	S = G/C	W = A/T	K = G/T	M = A/C
B = C/G/T	D = A/G/T	H = A/C/T	V = A/C/G	N = any base	

Table 8.2: Some Online Analysis Tools for Prediction of Post-Translational Protein Modifications, Protein Sorting, Localization Signals.

Online Tool	**URL**
CBS Prediction Servers (Center for Biological Sequence Analysis, Technical University of Denmark DTU)	http://www.cbs.dtu.dk/services/*
PSORT (Protein Sorting)	http://psort.hgc.jp/+
Gene Infinity	http://www.geneinfinity.org/sp/sp_proteinptmodifs.html++

Table 8.7: Some Online Tools for Transmembrane – Helix Prediction

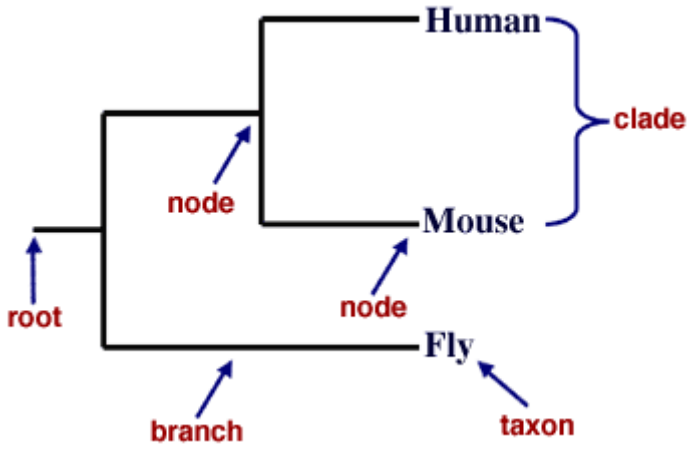
Online Tool	**Comments and URL**
TMHMM	Hidden-Markov-model-based[46] (http://www.cbs.dtu.dk/services/TMHMM/)
RHYTHM	Utilizes the structural information from ever-growing data sets and evolutionary information from conserved-sequence patterns in a representative data set of membrane proteins[47] (http://proteinformatics.charite.de/rhythm/)
Phobius	Hidden-Markov-model-based[48]

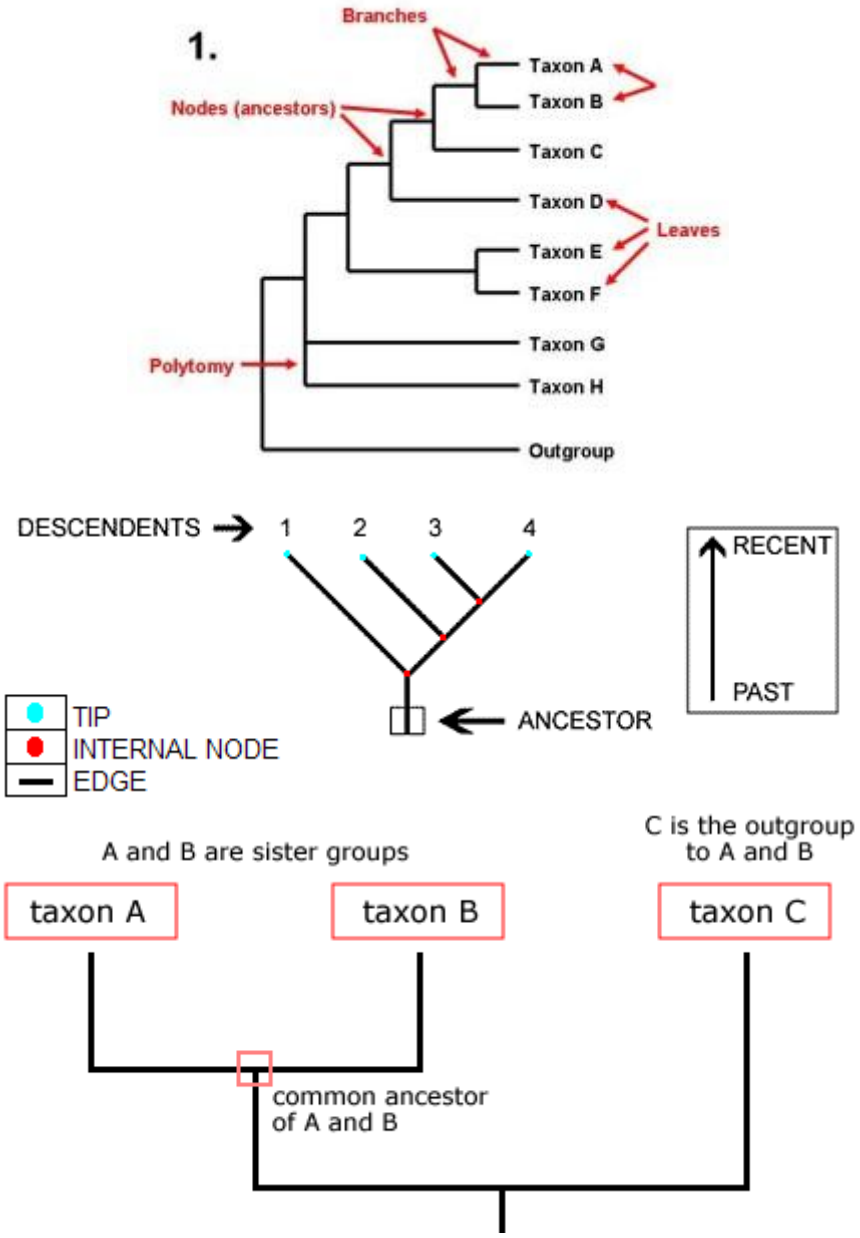
	(http://phobius.sbc.su.se/)
OCTOPUS	Artificial-neural-network-based[49] (http://octopus.cbr.su.se/)

Phylogeny refers to the evolutionary history of species. **Phylogenetics** is the study of phylogenies—that is, the study of the evolutionary relationships of species. **Phylogenetic analysis** is the means of estimating the evolutionary relationships.

The evolutionary relationship obtained from phylogenetic analysis is usually depicted as branching, treelike diagram—the **phylogenetic tree**.

A phylogenetic tree or evolutionary tree is a diagrammatic representation of the evolutionary relationships among various taxa (Figure 9.1 A-D). It is a branching diagram composed of **nodes and branches**. The branching pattern of a tree is called the topology of the tree.





The **nodes represent taxonomic units, such as species (or higher taxa), populations, genes, or proteins**.
A **branch is called an edge**, and **represents the time estimate of the evolutionary relationships among the taxonomic units**.

One branch can connect only two nodes.

In a phylogenetic tree, **the terminal nodes represent the operational taxonomic units (OTUs) or leaves (Fig B) . The OTUs are the actual objects**—such as the species, populations, or gene or protein sequences—being compared, **whereas the internal nodes**

represent hypothetical taxonomic units (HTUs). **An HTU is an inferred unit and it represents the last common ancestor (LCA) to the nodes arising from this point**.

Descendants (taxa) that split from the same node form sister groups, and a taxon that falls outside the cladea is called an outgroup.

For example, in Figure 9.1 B, **T2 and T3 are sister groups**, and **T1 is an outgroup to T2 and T3**.

Phylogenetic trees can be scaled or unscaled. In a scaled tree, the branch length is proportional to the amount of evolutionary divergence (e.g. the number of nucleotide substitutions) that has occurred along that branch. In an unscaled tree, the branch length is not proportional to the amount of evolutionary divergence, but usually the actual number is indicated somewhere on the branch.

Phylogenetic trees can be rooted (Figure 9.1 A and B) or unrooted (Figure 9.1 C). **A rooted tree has a node (the root) from which the rest of the tree diverges. This root is frequently referred to as the last universal common ancestor (LUCA) (FIG B),** from which the other taxonomic groups have descended and diverged over time. **In molecular phylogenetics, the LUCA and LCA are represented by DNA or protein sequences. Obtaining a rooted tree is ideal,** but **most phylogenetic-tree reconstruction algorithms produce unrooted trees**

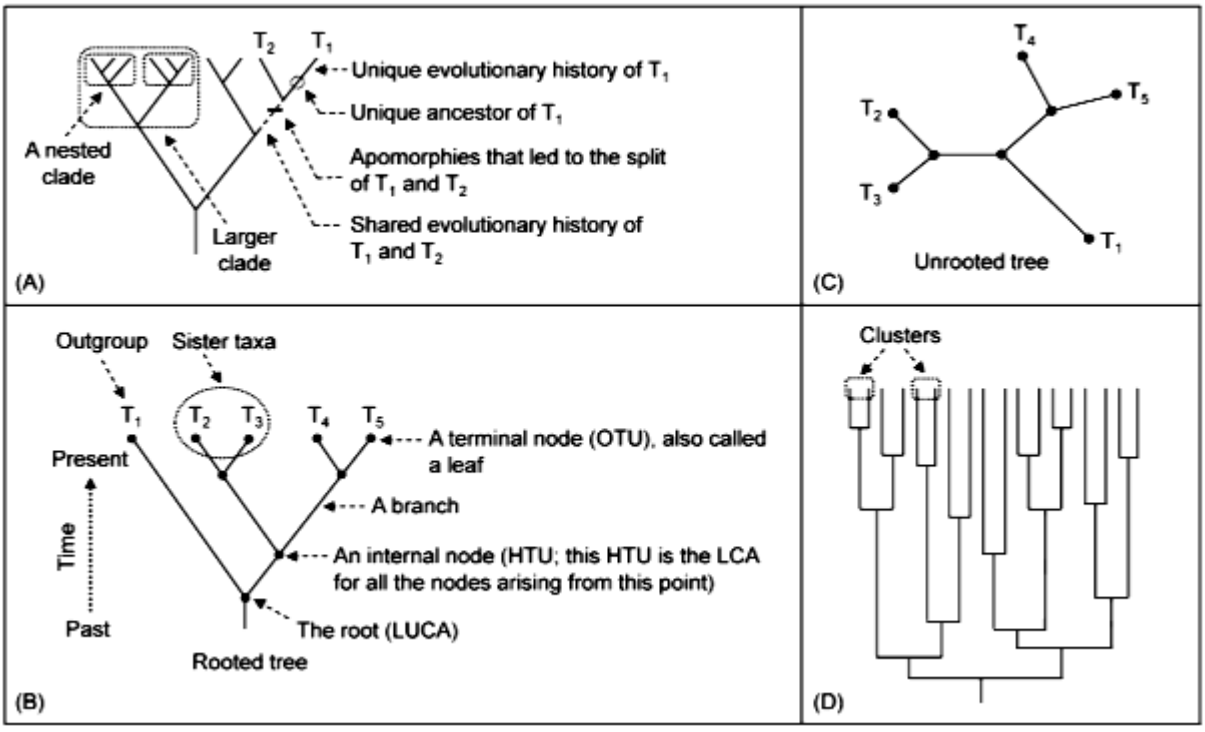


Figure 9.1: Different forms of presentation of the phylogenetic tree. The phylogenetic tree in D is a dendrogram derived from hierarchical clustering (see text). A, B, and D show rooted trees, while C shows an unrooted tree. taxa that share specific derived characters are grouped

into clades. (A) Smaller clades located within a larger clade are called nested clades. (B) The terminal nodes represent the operational taxonomic units, also called "leaves"; each terminal node could be a taxon (species or higher taxa), or a gene or protein sequence. The internal nodes represent hypothetical taxonomic units. An HTU represents the last common ancestor to the nodes arising from this point. Two descendants that split from the same node are called sister groups and a taxon that falls outside the clade is called an outgroup. Rooted trees have a node from which the rest of the tree diverges, frequently called the last universal common ancestor (LUCA).

[a]: Taxa that share specific derived characters are grouped more closely together than those who do not. The groups are called clades; each clade consists of an ancestor and all of its descendants.

Number of Rooted and Unrooted Trees

Number of unrooted trees for n taxa $N_u = (2n - 5)*(2n - 7)* \dots *3*1 = (2n - 5)!/[2^{n-3}*(n - 3)!]$

Number of rooted trees for n tax $N_r = (2n - 3)*(2n - 5)*(2n - 7)* \dots *3*1 = (2n - 3)!/[2^{n-2}*(n - 2)!]$

Note that the number of unrooted trees for n sequences is equal for the number of rooted trees for (n – 1) sequences.

Number of Taxa	**Number of unrooted trees**	**Number of rooted trees**
3	1	3
4	3	15
5	15	105
6	105	945
7	945	10395
8	10395	135135

Phylogenetics also provides the basis for comparative genomics, which is a more recent term that came into existence in the age of genomics.

In mathematics, a factorial is a function applied to natural numbers greater than zero. The symbol for the factorial function is an exclamation mark after a number, like this: 2! The usual definition of "n factorial" is "n! = n * (n – 1) * (n – 2) *…* 2 * 1," where 'n' is a positive integer.

Molecular Dynamics and Simulations

Molecualr Dynamics is a computational method which describes equilibrium and dynamics properties of a biological system. Molecular dynamics (MD) simulations represent the computer approach to statistical mechanics. As a counterpart to experiment, MD simulations are used to estimate equilibrium and dynamic properties of complex systems that cannot be calculated analytically. Representing the exciting interface between theory and experiment, MD simulations occupy a venerable position at the crossroads of mathematics, biology, chemistry, physics, and computer science. The static view of a biomolecule, as obtained from X-ray crystallography for example - while extremely valuable - is still insufficient for understanding a wide range of biological activity. It only provides an average, frozen view of a complex system. Certainly, molecules are live entities, with their constituent atoms continuously interacting among themselves and with their environment. Their dynarnic motions can explain the wide range of thermally-accessible states of a system and thereby connect structure and function. By following the dynarnics of a molecular system in space and time, we can obtain a rich amount of information concerning structural and dynarnic properties.

The molecular dynamics method was first introduced by Alder and Wainwright in the late 1950's (Alder and Wainwright, 1957,1959) to study the interactions of hard spheres. Many important insights concerning the behavior of simple liquids emerged from their studies.

The next major advance was in 1964, when Rahman carried out the first simulation using a realistic potential for liquid argon (Rahman, 1964). The first molecular dynamics simulation of a realistic system was done by Rahman and Stillinger in their simulation of liquid water in 1974 (Stillinger and Rahman, 1974). The first protein simulations appeared in 1977 with the simulation of the bovine pancreatic trypsin inhibitor (BPTI) (McCammon, et al, 1977).

Today in the literature, one routinely finds molecular dynamics simulations of solvated proteins, protein-DNA complexes as well as lipid systems addressing a variety of issues including the thermodynamics of ligand binding and the folding of small proteins. The number of simulation techniques has greatly expanded; there exist now many specialized techniques for particular problems, including mixed quantum mechanical classical simulations, that are being employed to study enzymatic reactions in the context of the full protein. Molecular dynamics simulation techniques are widely used in experimental procedures such as Xray crystallography and NMR structure determination.

Furthermore their use is no longer reserved to experts and many experimentalists use computer simulations nowadays as a tool to analyze or interpret their measurements whenever they are too complex to be described by simple analytical models. This is particularly true in the case of neutron scattering, as in this case the experimental observables correlate directly with the properties obtained in MD simulations and the spatial and time scales that can be measured match very well those amenable to computation. Such features, together with the availability of user-friendly and reliable software to perform this kind of calculation (e.g. Charmm , NAMD , Amber , Gromacs , Gromos , DL_POLY, etc.) and to visualize and

analyse their output (VMD, gOpenMol , nMoldyn, . . .), enable that today MD simulations are routinely used by many neutron users.

Even if the available software makes quite easy for a novice user to perform a MD simulation without needing lengthy training, it is advisable to acquire a minimum knowledge on the principles of the technique, as well as on the meaning of the terms and choices that one may encounter when using any MD program. The goal of the present chapter is precisely to give to the reader such basic knowledge. But if we want to study a particular system by means of a MD simulation, the requirement of a good program to perform the computation is a necessary but not sufficient condition. In addition we need a good model to represent the interatomic forces acting between the atoms composing the system. Ideally this can be done from first principles, solving the electronic structure for a particular configuration of the nuclei, and then calculating the resulting forces on each atom. Since the pioneering work of Car and Parrinello the development of *ab initio* MD (AIMD) simulations has grown steadily and today the use of the density functional theory (DFT) allows to treat systems of a reasonable size (several hundreds of atoms) and to achieve time scales of the order of hundreds of ps, so this is the preferred solution to deal with many problems of interest. However quite often the spatial and/or time-scales needed are prohibitively expensive for such ab initio methods. In such cases we are obliged to use a higher level of approximation and make recourse to empirical force field (FF) based methods. They allow to simulate systems containing hundreds of thousands of atoms during times of several nanoseconds or even microseconds. On the other hand the quality of a force field needs to be assessed experimentally. And here again neutrons play a particularly important role because while many different experimental results can be used to validate some of the FF parameters, the complementarity mentioned above makes neutron scattering a particularly useful tool in this validation procedure.

Statistical mechanics is the branch of physical sciences that studies macroscopic systems from a molecular point of view. The goal is to understand and to predict macroscopic phenomena from the properties of individual molecules making up the system. The system could range from a collection of solvent molecules to a solvated protein-DNA complex. In order to connect the macroscopic system to the microscopic system, time independent statistical averages are often introduced. We start this discussion by introducing a few definitions.

Definitions

The thermodynamic state of a system is usually defined by a small set of parameters, for example, the temperature, T, the pressure, P, and the number of particles, N. Other thermodynamic properties may be derived from the equations of state and other fundamental thermodynamic equations.

The **mechanical or microscopic state** of a system is defined by the atomic positions, q, and momenta, p; these can also be considered as coordinates in a multidimensional space called phase space. For a system of N particles, this space has 6N dimensions. A single point in phase space, denoted by Γ, describes the state of the system. An ensemble is a collection of points in phase space satisfying the conditions of a particular thermodynamic state. A molecular dynamics simulations generates a sequence of points in phase space as a function of time; these points belong to the same ensemble, and they correspond to the different

conformations of the system and their respective momenta. Several different ensembles are described below.

An **ensemble** is a collection of all possible systems which have different microscopic states but have an identical macroscopic or thermodynamic state. There exist different ensembles with different characteristics.

Microcanonical ensemble (NVE) : The thermodynamic state characterized by a fixed number of atoms, N, a fixed volume, V, and a fixed energy, E. This corresponds to an isolated system. *Canonical Ensemble (NVT)*: This is a collection of all systems whose thermodynamic state is characterized by a fixed number of atoms, N, a fixed volume, V, and a fixed temperature, T.

IsobaricIsothermal Ensemble (NPT): This ensemble is characterized by a fixed number of atoms, N, a fixed pressure, P, and a fixed temperature, T. Grand canonical Ensemble (μVT): The thermodynamic state for this ensemble is characterized by a fixed chemical potential, μ, a fixed volume, V, and a fixed temperature, T.

FORCE FIELDS

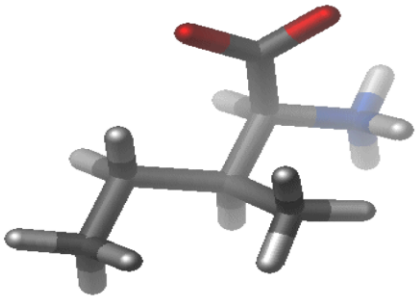
A **force field** is a mathematical expression describing the dependence of the energy of a system on the coordinates of its particles. It consists of an analytical form of the interatomic potential energy, and a set of parameters entering into this form. The parameters are typically obtained either from ab initio or semi-empirical quantum mechanical calculations or by fitting to experimental data such as neutron, X-ray and electron diffraction, NMR, infrared, Raman and neutron spectroscopy, etc. Molecules are simply defined as a set of atoms that is held together by simple elastic (harmonic) forces and the FF replaces the true potential with a simplified model valid in the region being simulated. Ideally it must be simple enough to be evaluated quickly, but sufficiently detailed to reproduce the properties of interest of the system studied. There are many force fields available in the literature, having different degrees of complexity, and oriented to treat different kinds of systems.

The Effective molecular energy can be expressed as a sum of potentials derived simple physical forces

$$V = E_{bonded} + E_{nonbonded}$$

$$E_{bonded} = E_{bond-strech} + E_{angle-bend} + E_{rotate-along-bond}$$

$$E_{nonbonded} = E_{van-der-waals} + E_{electrostatic}$$



However a typical expression for a FF may look like this:

$$U = \sum_{bonds} \frac{1}{2} k_b \left(r - r_O\right) + \sum_{angles} \frac{1}{2} k_a \left(\theta - \theta_O\right)^2 + \sum_{torsions} \frac{V_n}{2} \left[1 + \cos\left(n\phi - \delta\right)\right]$$

$$+ \sum_{improper} V_{imp} + \sum_{LJ} 4 \in_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^{6}}{r_{ij}^{6}} \right) + \sum_{elec} \frac{q_i q_j}{r_{ij}}$$

where the first four terms refer to intramolecular or local contributions to the total energy (bond stretching, angle bending, and dihedral and improper torsions), and the last two terms serve to describe the repulsive and Van der Waals interactions (in this case by means of a 12-6 Lennard-Jones potential) and the Coulombic interactions.

Molecuar Dynamics

An MD simulation is a technique to produce a dynamical trajectory for a system composed of N particles by integrating Newton's equations of motion.We need a set of initial conditions (positions and velocities of each particle), a good model to represent the forces acting between the particles (either from electronic structure calculations or using the empirical force fields presented in the previous section), and to define the boundary conditions to be employed. Then we need to solve the classical equation of motion:

$$m_i \frac{d^2 r_i}{dt^2} = f_i = -\frac{\partial}{\partial r_i} U\left(r_1, r_2, ..., r_N\right)$$

where U(r1, r2, . . . , rN) is the potential energy depending on the coordinates of the N particles. This is a system of N coupled second order non linear differential equations that cannot be solved exactly, so equation above has to be solved numerically step by step using an appropriate integration algorithm.

SIMULATION

Molecular dynamics (MD) simulation, first developed in the late 70s, has advanced from simulating several hundreds of atoms to systems with biological relevance, including entire proteins in solution with explicit solvent representations, membrane embedded proteins, or large macromolecular complexes like nucleosomes or ribosomes. Simulation of systems having ~50,000–100,000 atoms are now routine, and simulations of approximately 500,000 atoms are common when the appropriate computer facilities are available. This remarkable improvement is in large part a consequence of the use of high performance computing (HPC), and the simplicity of the basic MD algorithm

We can consider that there are four different steps in a MD simulation:

1. **Setting up the system**: This stage includes most of the aspects mentioned in the preceding sections. We have to create a reasonable configuration of the system we are interested in, choose a valid force field for this system, select a cut-off radius and the method to treat the electrostatic interactions if the system contains partial charges, choose an integration algorithm (if the MD code we use allows it) and a correct time step, and select the working ensemble (NVE, NVT or NPT). At this point we have to be sure that the size of the system is enough to explore the physics of interest.

2. **Equilibration**: Normally the initial configuration will not be representative of the conditions we want to explore. For example, if it comes from a previous MD simulation we may want to simulate the system at a different temperature. If the initial positions were determined randomly we may have some atoms that are too close or a lack of short-range correlations which is not real. On the other hand, if we start from an ordered configuration and we are interested in the liquid, we need to melt the system and guarantee that the long-range order has completely disappeared. This step usually requires the use of a NPT simulation in order to allow the system to achieve the equilibrium density corresponding to the desired pressure and temperature. However the choice of the thermostat and barostat is not critical, as this part of the simulation will not be used to compute any property. It is important to ensure that we have achieved an equilibrated state, so during this phase we should follow the pressure, the density and the different energy components, which at equilibrium should fluctuate around some average value without showing any drift. If we have started from a lattice configuration and we are melting it, we can also check the evolution of a translational and/or rotational order parameter. Finally we can verify that some standard properties such as the radial distribution function or the mean square displacement are independent of the segment of the trajectory employed to compute them, but we have to be aware that this will be the case only when the segment is long enough so that the ergodic hypothesis holds.

3. **Production**: Once we have attained an equilibrium situation at the desired temperature and pressure, we can perform the production run. It may be convenient to work in the NVE ensemble to avoid any influence from the thermostat or the barostat, as we saw before. Nevertheless in very long simulations the drift in temperature originated mainly from truncation effects can be embarrassing, so we may need to apply any of the thermostats described above. The pressure should oscillate around the average value achieved during the equilibration, so normally there is no need to use any barostat. Thus we can work with a fixed cell and avoid the inconvenience of dealing with volume fluctuations during the analysis of the trajectory. A clear exception here is in the case where we are interested in simulating a phase transition. We also have to decide how long to run the simulation. We need to run it for a time much longer than the relaxation time of the property of interest. But the simulation time should also be sufficiently long to ensure the validity of the ergodic hypothesis, i.e. that the time average of a given property coincides with the ensemble average. While this condition is true in most cases (such as when simulating a standard liquid or crystal), some systems such as glasses are not ergodic, so the results will depend on the history of the system. Similarly, if we study a large biomolecule that can adopt different conformations, the simulation time should be enough to allow the macromolecule to explore all the possible

configurations. Otherwise the results obtained (e.g. atomic displacements) may correspond to one of the global conformations of the macromolecule, but not to the real situation.

4. **Analysis**: Finally the simulated trajectory must be analysed to extract the desired properties. We have access to the atomic positions, velocities and even forces as a function of time, so any statistical mechanical property that can be expressed in terms of those variables can be computed. In the following section we present some examples of the different properties that can be computed in a MD simulation, while a more detailed account can be found in this collection.

Introduction to Molecular dynamics of Proteins and Nucleic acids

The study of the macromolecular structure is a key point in the understanding of biology. Biological function is based on molecular interactions, and these are a consequence of macromolecular structures. Since initial structure determinations in the 50s, both in the protein and in the nucleic acid worlds, the increase in the knowledge of how macromolecular structures are built has been continuous. At present, protein data bank (PDB)1 holds more than 110,000 entries, including more than 100,000 proteins, 2,800 nucleic acids, alone or forming complexes, and approximately 20,000 small molecules complexed to macromolecules. Molecular recognition rules as defined by such structural knowledge powers the understanding of basic biological phenomena, like enzyme mechanisms and regulation, transport across membranes, the building of large structures like ribosomes, or viral capsids, or how DNA is read and transcription is controlled. The study and prediction of protein–protein interaction networks is one of the growing fields in modern systems biology. On a more practical note, protein three-dimensional (3D) structures are the basis for structure-based drug design. The simple visual analysis of 3D structures of protein or nucleic acids, as obtained from the experiments, has driven large number of successful studies in biochemistry. However, despite their enormous utility, structures stored at the PDB provide only a partial view of 3D structure. Both protein and nucleic acids are flexible entities, and dynamics can play a key role in their functionality. Proteins undergo significant conformational changes while performing their function. As a rule, any complex made by any protein implies some structural rearrangement. This can be easily checked just by comparing a series of PDB entries that just differ in a small ligand bound to a given protein.
Binding to ligands (allosteric regulators), or simply protein concentration, or crowding, may switch stabilities among conformations and trigger the shape transition. Additionally, some features of protein function can be understood only when dynamic properties are taken into account. For instance, diffusion of small substrates through heme-dependent enzyme molecules requires the transient appearance of channels in the protein structure. Also, cavities have transient phenomena that in some cases can only be revealed or analyzed following its dynamics.

In the case of nucleic acids, conformational changes are even more complex. Standard B-DNA has a relatively simple structure in comparison with protein or complex RNAs; however, it is an extremely plastic molecule that undergoes large conformational changes to adapt to its interaction partners. Binding of transcription factors to DNA, for example, is not

only dependent on DNA sequence recognition, but also a direct consequence of the ability of the DNA molecule to adapt to the protein surface.